End-to-End Process · Plan-to-Make (P2M)

PCF 4.1.2 Manage Demand for Products · PCF 4.1.3 Create Materials Plan · PCF 4.1.4 Create and Manage Master Production Schedule · PCF 4.1.5 Plan Distribution Requirements

View 1 — HITL Design

Pattern Type Legend

Decision	Knowledge	Document	Transaction	Exception
80% H at M3	65% H	40% H	35% H	82% H

P2M End-to-End Process — HITL Design · PCF-Derived Process Ownership

Demand Planning	Materials Planning	Production Scheduling	Distribution Planning
PCF 4.1.2	*PCF 4.1.3*	*PCF 4.1.4*	*PCF 4.1.5*

HITL Design Insight

HITL Design Insight — P2M

Manage Demand for Products (4.1.2, 65% Human at M3) covers developing baseline statistical demand forecasts, collaborating with customers and sales on consensus forecasts, determining available-to-promise, and monitoring and measuring forecast accuracy. Classified as a Knowledge process: demand consensus and forecast revision are judgment-heavy, cross-functional activities even though baseline forecasting itself is statistically generated. **Create Materials Plan** (4.1.3, 65% Human at M3) covers generating the unconstrained materials plan from the demand forecast, collaborating with suppliers and contract manufacturers on capacity, identifying critical materials and supplier constraints, and generating the constrained plan that balances supply against demand. Also classified Knowledge-dominant: constraint-balancing and supplier collaboration require ongoing judgment, even as algorithmic optimization handles the underlying plan generation. **Create and Manage Master Production Schedule** (4.1.4, 35% Human at M3) covers modeling the production network for simulation and optimization, and creating and maintaining the master production schedule. Classified as a Transaction process: once the planning model is built, schedule generation and maintenance follow a rules-based system-of-record cadence. **Plan Distribution Requirements** (4.1.5, 35% Human at M3) covers maintaining distribution master data, calculating finished-goods and storage requirements at destination, managing collaborative replenishment planning, and calculating and optimizing dispatch and load plans. Also classified Transaction-dominant: requirements calculation and load/dispatch optimization run across many SKUs and destinations on an algorithmic, rules-based basis. P2M aggregate at M3: **50.0% Human / 50.0% AI** — an even split reflecting P2M's blended character: demand and materials planning remain Knowledge-dominant, judgment-heavy cycles, while production scheduling and distribution planning are Transaction-dominant, algorithmic optimization run at scale, giving the module a mid-range M1→M3 gap (-18.2pt) between L2O's judgment-heavy and P2P's transaction-heavy profiles.

HITL Impact Analysis

H% = Human involvement at each maturity level · M3 = HITL Design Intent · ★ = Design target

Process Step (L3)	PCF	Pattern	M1 H%	M3 H% ★	Δ M1→M3	Opportunity
P2M End-to-End Process 4.1.2, 4.1.3, 4.1.4, 4.1.5	—	Mixed	68.2%	50.0%	-18.2%	Moderate
Demand Planning (PCF 4.1.2)						

End-to-End Process · Plan-to-Make (P2M)

PCF 4.1.2 Manage Demand for Products · PCF 4.1.3 Create Materials Plan · PCF 4.1.4 Master Production Schedule · PCF 4.1.5 Plan Distribution Requirements

Manage Demand for Products	4.1.2	**Knowledge**	78%	**65%**	**-13%**	Moderate
Materials Planning (PCF 4.1.3)						
Create Materials Plan	4.1.3	**Knowledge**	75%	**65%**	**-10%**	Moderate
Production Scheduling (PCF 4.1.4)						
Create and Manage Master Production Schedule	4.1.4	**Transaction**	60%	**35%**	**-25%**	High
Distribution Planning (PCF 4.1.5)						
Plan Distribution Requirements	4.1.5	**Transaction**	60%	**35%**	**-25%**	High

AI Capability Map — P2M

PCF 4.1.2 Demand Planning

Baseline statistical forecast generation	KNOWLEDGE
Demand consensus synthesis across functions	KNOWLEDGE
Available-to-promise calculation	TRANSACTION
Forecast accuracy monitoring & variance flagging	KNOWLEDGE

M1: **78%** M3: **65%** M5: **50%**

PCF 4.1.3 Materials Planning

Unconstrained plan generation	TRANSACTION
Supplier & contract-manufacturer capacity collaboration	KNOWLEDGE
Critical-material & constraint identification	KNOWLEDGE
Constrained plan generation & balancing	KNOWLEDGE

M1: **75%** M3: **65%** M5: **48%**

PCF 4.1.4 Production Scheduling

Production network modeling & simulation	KNOWLEDGE
Master production schedule generation	TRANSACTION
Schedule maintenance & adjustment	TRANSACTION

M1: **60%** M3: **35%** M5: **20%**

PCF 4.1.5 Distribution Planning

Finished-goods & storage requirement calculation	TRANSACTION
Collaborative replenishment planning	KNOWLEDGE
Dispatch & load plan optimization	TRANSACTION
Capacity utilization & cost-of-supply management	TRANSACTION

M1: **60%** M3: **35%** M5: **20%**